

Sunspot Number V2.0 Through Solar Cycle 25

A Long-term Multi-Proxy Stability Analysis

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The Sunspot Number is the longest direct index of solar activity.

- SN V2.0 is the modern reference scale released after major recalibration.
- It is observer- and network-based: counting rules, coverage, and observer scaling all matter.
- Independent proxies provide external checks on whether SN behaves consistently.
- A disagreement with one proxy is not enough to claim SN drift.

Context: Clette & Lefèvre (2016); Clette et al. (2023); SILSO Sunspot Number V2.0.

Does SN V2.0 show coherent drift relative to independent solar-activity records?

- SN is the reference under test.
- Proxies sample different layers or observing chains.
- Strong evidence requires sustained, multi-proxy residual structure.

Interpretive rule: coherent multi-proxy behavior matters more than any single-proxy excursion.

Strategy: A Four-Step Stability Test

1

Align monthly data

Common timestamps,
monthly means, canonical
smoothing.

2

Fit proxy from SN

Each proxy gets its own
SN-proxy relation.

3

Reconstruct proxy

Ask what the proxy should
look like if SN explains it.

4

Inspect coherence

Look for persistent
departures shared by several
proxies.

Dataset: Reference and Photospheric Records

Series	Span used	Meaning	Role here	Source
SN V2.0	1818–present	Observer/network sunspot index	Reference scale under test	SILSO; Clette & Lefèvre
Sunspot area	1874–present	Photospheric area coverage	Long geometric proxy	Mandal et al. 2020
LOS magnetic field	1996–present	Photospheric magnetic signal	Magnetic-field context; shorter span	Scherrer 1995; Schou 2012

The photospheric records are closest to the spot-counting problem, but they still measure different quantities: counts, area, and magnetic flux.

Dataset: Radiative and Observer-Chain Checks

Series	Span used	Meaning	Role here	Source
F10.7	1947–present	10.7 cm radio flux	Upper-atmosphere activity proxy	Tapping 2013; Clette 2021
Mg II	1978–present	UV core-to-wing index	Chromospheric proxy	Viereck; de Toma; Bremen
AAVSO R_a	1945–present	Independent sunspot counts	Observer-chain comparison	AAVSO: Shapley–Hossfield

These records should follow the solar cycle, but their response to sunspots need not be linear or identical across cycle phase.

Method: Promote One Diagnostic

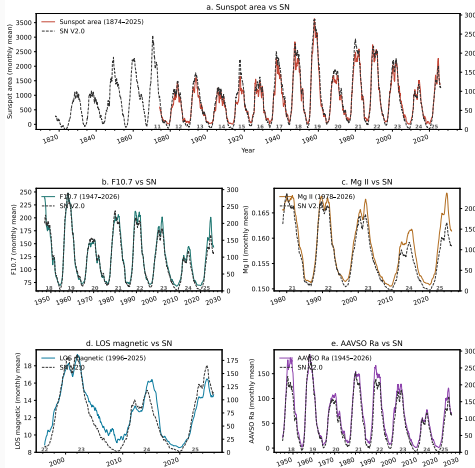
β measures the fractional proxy residual after reconstructing the proxy from SN.

$$\beta(t) = \frac{X_{\text{obs}}(t) - X_{\text{recon}}(t)}{X_{\text{obs}}(t)}$$

- X_{obs} is the measured proxy.
- X_{recon} is the proxy expected from the fitted SN–proxy relation.
- β near zero means the proxy behaves as expected from SN.
- Sustained non-zero β means the relation is unusual.

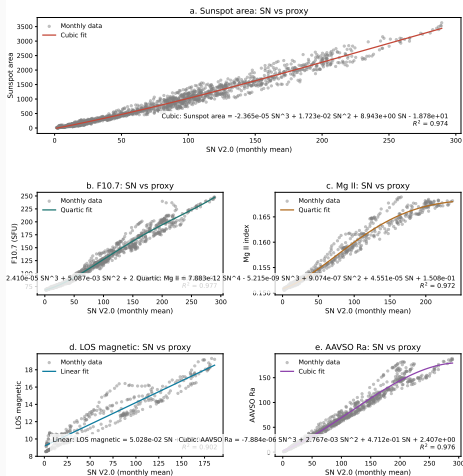
This follows the broad literature practice of comparing departures from expected SN–proxy relations; the exact β definition is the diagnostic used in this workflow.

Result 1: All Proxies Follow the Solar Cycle



- The main cycle phasing is shared across records.
- Different start dates define different overlap intervals.
- This establishes comparability; it is not yet a stability test.

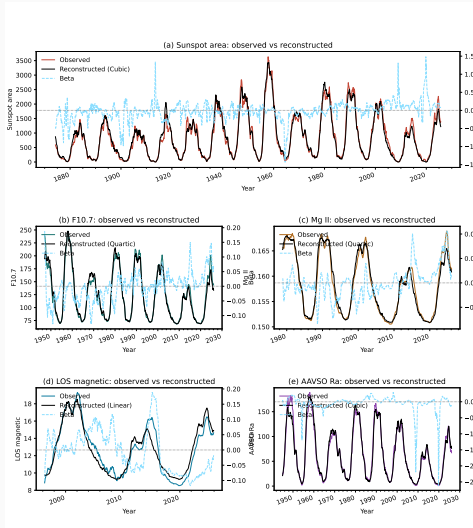
Result 2: Nonlinear Relations Matter



- Proxy response to SN is not always linear.
- F10.7 is especially sensitive to averaging and low-activity curvature.
- Selected families: area cubic, F10.7 quartic, Mg II quartic, LOS linear, R_a cubic.

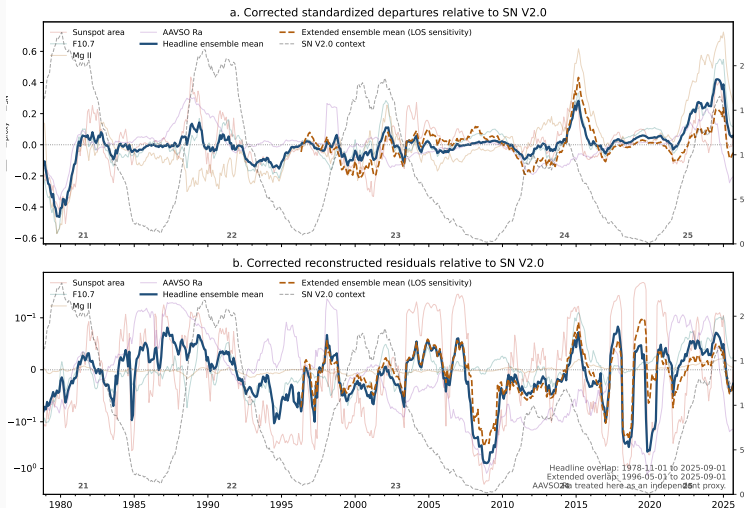
Context: Tapping & Morgan (2017); Clette (2021); Okoh & Okoro (2020).

Result 3: Beta Shows When Relations Change



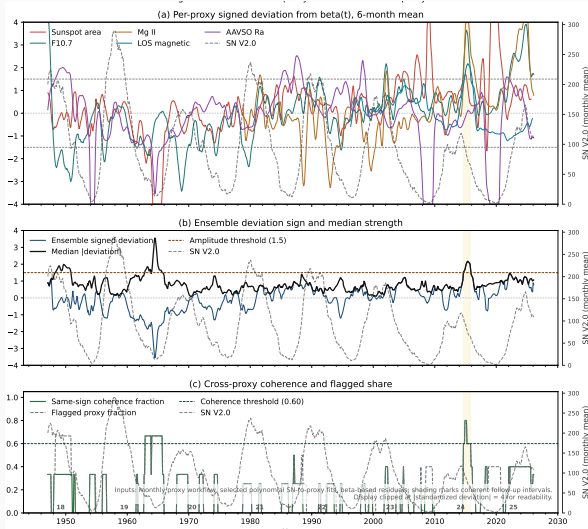
- β is the main residual diagnostic.
- Month-scale excursions are expected.
- Sustained structure is more important than isolated noise.
- Same-sign behavior across proxies is the signal to follow.

Result 4: The Common-Overlap Ensemble Stays Small



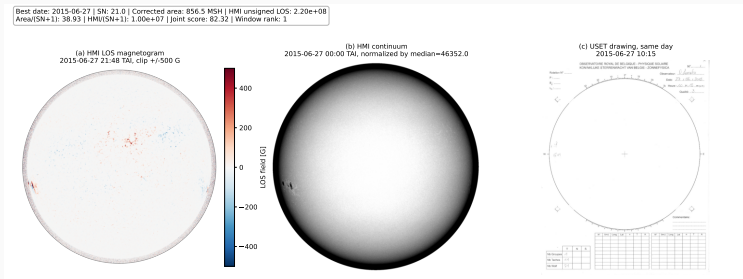
- SN is the sole reference.
- AAVSO R_a is an independent proxy, not a second reference.
- Headline overlap: 1978-11 to 2025-09.
- Mean β : -0.0258 headline; -0.0151 with LOS.
- Main reading: small mean with proxy-dependent structure.

Result 5: Deviation Windows Are Follow-Up Maps



- Coherent interval: 2014-09 to 2015-11.
- Peak month: 2015-05.
- Peak support: F10.7, LOS magnetic, and Mg II.
- This identifies a target window; it does not prove SN drift.

Result 6: Modern Follow-Up Is Image-Based



- Shown: strongest daily follow-up target, 2015-06-27.
- Seven candidate windows have HMI magnetogram, HMI continuum, and USET panels.
- Purpose: test whether proxy excess reflects solar structures or count-production effects.

Modern area context: Meadows (2025).
HMI instrument context: Schou et al. (2012).

What the Results Mean

- The proxies broadly validate the solar-cycle structure of SN V2.0.
- Some intervals show organized residual behavior.
- The key modern follow-up interval is 2014–2015.
- Proxy-dependent departures are expected because proxies sample different solar layers and observing processes.
- The evidence should not be phrased as proof of a simple SN failure.

Conservative reading: stable overall, with targeted intervals that deserve closer physical and observer-chain inspection.

Physical-context caution: Tapping & Morgan (2017); Clette (2021); Mursula et al. (2024).

- SN V2.0 is tested as the reference, not treated as one proxy among many.
- Independent proxies show strong cycle-scale agreement with SN.
- β residuals provide a clear, interpretable stability diagnostic.
- Coherent residual windows provide targeted follow-up intervals.
- The strongest next step is image-level and observer-chain inspection of flagged modern intervals.

- Inspect flagged 2014–2015 dates using HMI magnetograms, HMI continuum images, and USET drawings.
- Compare same-day and adjacent-day drawings when exact USET dates are missing.
- Test robustness to smoothing window, fit family, and proxy subset.
- Separate solar-physical proxy divergence from possible SN production effects.
- Document follow-up results in the manuscript and supplementary tables.

References: Sunspot Number and Proxy Context

- Clette, F. & Lefèvre, L. 2016, *Solar Physics*, 291, 2629–2651.
- Clette, F. et al. 2023, *Solar Physics*, 298, 44.
- SILSO Sunspot Number V2.0, World Data Center SILSO, Royal Observatory of Belgium, DOI: 10.24414/qnza-ac80.
- Tapping, K. F. 2013, *Space Weather*, 11, 394–406.
- Tapping, K. & Morgan, C. 2017, *Solar Physics*, 292, 73.
- Clette, F. 2021, *Journal of Space Weather and Space Climate*, 11, 2.
- Okoh, D. & Okoro, E. 2020, *Solar Physics*, 295, 1.
- Mursula, K. et al. 2024, *Astronomy & Astrophysics*, 685, A170.

References: Data Records Used Here

- Mandal, S. et al. 2020, *Astronomy & Astrophysics*, 640, A78.
- Meadows, P. J. 2025, *RAS Techniques and Instruments*, 4, rzaf024.
- Viereck, R. A. et al. 1999, *Geophysical Research Letters*, 26, 2805–2808.
- de Toma, G. et al. 1997, *Journal of Geophysical Research*, 102, 2597–2606.
- Weber, M. and IUP Bremen UVSAT team, Composite Mg II index.
- Scherrer, P. H. et al. 1995, *Solar Physics*, 162, 129–188.
- Schou, J. et al. 2012, *Solar Physics*, 275, 229–259.
- Shapley (1949), Taylor (1985), Schaefer (1997), and Hossfield (2002) for AAVSO solar-observing context.

Questions?